

Unit N°1 - Programming design and computer languages

SPEAKING. Group work. In groups of four discuss what you think *programming* is and write a definition.

VOCABULARY. Steps in programming. Match the words (1-5) to their definitions (a-e)

1. flowchart
 2. source code
 3. compiler
 4. machine code
 5. debugging
-
- a. Program instructions written in a particular computer language.
 - b. The techniques of detecting and correcting errors (or bugs) which may occur in programs.
 - c. A diagram representing the successive logical steps in a program.
 - d. A special program which converts the source program into machine code - the only language understood by the processor.
 - e. The basic instructions understood by computers; it consists of 1s and 0s (binary code).

LISTENING.

a) Read and put these steps into the correct order according to your knowledge. Then, listen to Andrea Finch, a software developer, talking to a group of students on a training course about how a program is written and check your answers. (Infotech - track 29)

- ___ Write instructions in a programming language.
- ___ Prepare documentation.
- ___ Understand the problem and plan the solution.
- ___ Make a flowchart of the program.
- ___ Compile the program.
- ___ Test and debug the program.

b) Listen again and make detailed notes. Then share them with the whole class.

READING COMPREHENSION

a) Read the text. How many high-level languages are mentioned?

Computer languages

Unfortunately for us, computers can't understand spoken English or any other natural language. The only language they can understand directly is **machine code**, which consists of 1s and 0s (binary code).

Machine code is too difficult to write. For this reason, we use symbolic languages to communicate instructions to the computer. For example, **assembly languages** use abbreviations such as ADD, SUB, MPY to represent instructions. The program is then translated into machine code by a piece of software called an **assembler**. Machine code and assembly languages are called **low-level languages** because they are closer to the hardware. They are quite complex and restricted to particular machines. To make the programs easier to write, and to overcome the problem of intercommunication between different types of computer, software developers designed **high-level languages**, which are closer to the English language. Here are some examples:

FORTRAN was developed by IBM in 1954 and is still used for scientific and engineering applications.

COBOL (**Common Business Oriented Language**) was developed in 1959 and is mainly used for business applications.

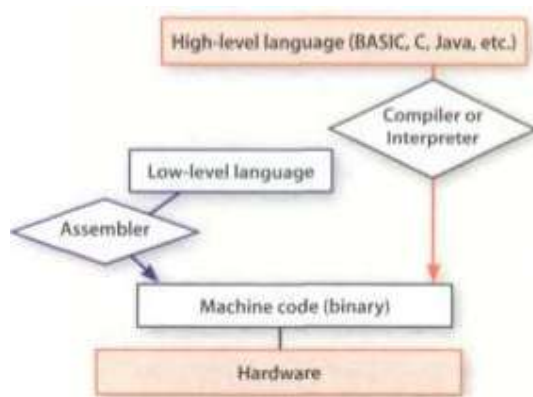
BASIC was developed in the 1960s and was widely used in microcomputer programming because it was easy to learn. **Visual BASIC** is a modern version of the old BASIC language, used to build graphical elements such as buttons and windows in Windows programs.

PASCAL was created in 1971. It is used in universities to teach the fundamentals of programming.

C was developed in the 1980s at AT&T. It is used to write system software, graphics and commercial applications. **C++** is a version of C which incorporates object-oriented programming: the programmer concentrates on particular things (a piece of text, a graphic or a table, etc.) and gives each object functions which can be altered without changing the entire program. For example, to add a new graphics format, the programmer needs to rework just the graphics object. This makes programs easier to modify.

Java was designed by Sun in 1995 to run on the Web. Java applets provide animation and interactive features on web pages.

Programs written in high-level languages must be translated into machine code by a **compiler** or an **interpreter**. A compiler translates the source code into **object code** - that is, it converts the entire program into machine code in one go. On the other hand, an interpreter translates the source code line by line as the program is running.



It is important not to confuse **programming languages** with **markup languages**, used to create web documents. Markup languages use instructions, known as **markup tags**, to format and link text files.

Some examples include:

HTML, which allows us to describe how information will be displayed on web pages.

XML, which stands for **EX**tensible **M**arkup **L**anguage. While HTML uses pre-defined tags, XML enables us to define our own tags: it is not limited by a fixed set of tags.

VoiceXML, which makes Web content accessible via voice and phone. VoiceXML is used to create voice applications that run on the phone, whereas HTML is used to create visual applications (for example, web pages).

```
<xml>
< name> Andrea Finch </name>
< homework> Write a paragraph describing
the C language </homework>
</xml>
```

In this XML example we have created two new tags: <name> and <homework>

b) Read the text again and answer these questions.

1 Do computers understand human languages? Why?/Why not?

2 What is the function of an assembler?

3 Why did software developers design high-level languages?

4 Which language is used to teach programming techniques?

5 What is the difference between a compiler and an interpreter?

6 Why are HTML and VoiceXML called markup languages?

c) Complete these sentences with a computer language from the text.

1 _____ allows us to create our own tags to describe our data better. We aren't

constrained by a pre-defined set of tags the way we are with HTML.

2 IBM developed _____ in the 1950s. It was the first high-level language in data processing.

3 _____ applets are small programs that run automatically on web pages and let you watch animated characters, play games, etc.

4 _____ is the HTML of the voice web. Instead of using a web browser and a keyboard, you interact with a voice browser by listening to pre-recorded audio output and sending audio input through a telephone.

5 This language is widely used in the business community. For example, the statement ADD VAT to NET-PRICE could be used in a _____ program.

WORD BUILDING. Look at the words in the boxes. Are they nouns, verbs or adjectives? Write n, v or adj next to each word. There may be more than one possible answer. Complete the sentences with words from the boxes.

program_____	programmers_____	programming_____	programmable_____
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1 _____ is the process of writing a program using a computer language.

2 A computer _____ is a set of instructions that tells the computer how to do a specific task.

3 Most computer _____ make a plan of the program before they write it.

4 A _____ keyboard allows the user to configure the layout and meaning of the keys.

compile_____	compiler_____	compilation_____
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5 Programs written in a high-level language require _____ - that is, translation into machine code, the language understood by the processor.

6 A source program is converted into machine code by software called a _____.

7 Programmers usually _____ their programs to generate an object program and diagnose possible errors.

bug_____	debug_____	debugger_____	debugging_____
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- 8 Any error or malfunction of a computer program is known as a _____.
- 9 A _____ is a program used to test and _____ other programs.
- 10 The process of going through the code to identify the cause of errors and fixing them is called _____.

GRAMMAR. the infinitive

Reflect upon grammar. Look back at the text “Computer languages” and underline all the “to + verb” that you can find. Then choose the correct option(s) to complete the sentence.

- We use “to” + an infinitive verb...
 - a) to express the purpose of something.
 - b) after adjectives.
 - c) as the main verb of a sentence.

The infinitive

The infinitive with to is used in the following ways:

- To express purpose

*We use symbolic languages **to communicate** instructions to the computer.*

*We use symbolic languages **in order to communicate** instructions to the computer.*

*We use symbolic languages **for to communicate** instructions to the computer.*

- After adjectives

*BASIC was widely used in the past because it was **easy to learn**.*

*Machine code is too **difficult to write** (= not easy enough to write)*

- After certain verbs (e.g. **afford, demand, plan, agree, expect, promise, appear, hope, refuse, arrange, learn, try, decide, manage**)

*A lot of companies are now **trying to develop** voice applications for web access.*

- After the object of certain verbs (e.g. **advise, encourage, allow, expect, tell, ask, invite, want, enable, order, warn**)

*HTML **allows us to describe** how information will be displayed on web pages.*

The bare infinitive (without to) is used in the following ways:

- After **modal verbs** (e.g. **can, could, may, might, will, would, must, should**)

*Unfortunately, computers **can't understand** spoken English.*

*High-level languages **must be** translated into machine code.*

- After the object with the verbs **make** and **let**

*Programs **make** computers **perform** specific tasks.*

LANGUAGE WORK: the infinitive

After dealing with the theory, make sentences using these prompts.

1 not easy / write instructions in COBOL

It's not easy to write instructions in COBOL

2 expensive / set up a data-processing area

3 advisable / test the programs under different conditions

4 unusual / write a program that works correctly the first time it's tested

5 important / use a good debugger to fix errors

6 easy / learn Visual BASIC

Graphics and design

SPEAKING. Discuss in pairs. As a programmer, is it important to know about graphic programs? If not, who would be in charge of the graphics?

LISTENING.

a) Listen to an extract from an online tutorial about graphics programs and answer these questions. Remember to focus on key words and write them down to answer the questions. (Infotech. Track 25)

1 What is a toolbox in graphics software?

2 What are graphics primitives?

3 What sort of attributes, or characteristics, can be used in graphical objects?

4 What does translation mean?

b) Listen again and complete this extract from the web version of the tutorial.

Graphics programs usually have a toolbox - a collection of drawing and (1) _____ tools that enable you to type, (2) _____ draw, paint, edit, move, and view images on the computer.

The basic shapes which are used to (3) _____ graphical objects are called primitives. These are usually geometric, such as lines between two points, arcs, circles, polygons, ellipses and even text. Furthermore, you can specify the attributes of each primitive, such as its colour, line type, fill area, interior style and so on.

The various tools in a toolbox usually appear together as pop-up icons in a menu or palette. To use one, you activate it by (4) _____ on it. For example, if you want to (5) _____ a rectangle, you activate the rectangle tool, and the pop-up options give you the possibility of (6) _____ rectangles with square or rounded corners.

You can transform an object by translating, (7) _____ or scaling it. Translation means moving an object to a different location. Rotation is (8) _____ the object around an axis. For example, you may need to rotate an object 90 or 180 degrees to fit the drawing. (9) _____ is making the object larger or smaller.

GRAMMAR. The -ing form

Reflect upon grammar. Read the text above (ex b) again and underline all the words that contain an *-ing suffix*. Are they used as verbs, adjectives or nouns? Provide examples for your answers.

The -ing form

We use the -ing form in three ways:

1) **Rendering** includes **lighting** and **shading**.

Rendering is a gerund acting as the subject of the sentence. **Lighting** and **shading** are also gerunds, but acting as the object of the sentence.

- A **gerund** is a word that is formed by a verb + -ing suffix used in a place where we traditionally would find a noun.
- They can be the subject or object of a sentence or they can be found after a preposition.
Designers start a project **by making** a wireframe.
- Some verbs are always followed by gerunds (e.g. **avoid, fancy, finish, give up, hate, imagine, involve, keep, look forward to, mind, suggest, enjoy, love, etc**)

2) They use special applets to create **amazing** fractals.

Amazing is an adjective.

3) We are **designing** a new graphic on this computer.

Designing is a *present participle*. Present participles are used in continuous tenses and reduced relative clauses.

LANGUAGE WORK. the -ing form.

a) After dealing with the theory, decide if the -ing forms in these sentences are gerunds, present participles or adjectives. Write **g**, **pp** or **a**.

- 1 PCs generate graphics by performing mathematical calculations on data. ____
- 2 Businesspeople use graphics to make information more interesting visually. ____
- 3 Graphs and diagrams can be more effective ways of communicating with clients than lists of figures. ____
- 4 She is designing a logo for the company. ____
- 5 If you need to make a presentation, I suggest using PowerPoint. ____
- 6 The Internet is a network linking other networks. ____

b) Correct the mistakes in these sentences. There are seven mistakes in total.

- 1 Computer animation is the process of create objects which move across the screen.
- 2 Texturing involves add paint, colour and filters to drawings and designs.
- 3 You can open the colour palette by click on the corresponding icon.
- 4 CAD programs are very fast at to perform drawing functions.
- 5 A lot of time and money is saved by test a car design before to make the product.
- 6 To render refers to the techniques used to make realistic images.

c) The infinitive or the -ing form? Choose the correct words (a-c) to complete these sentences.

1 We use high-level languages because machine code is too difficult _____, understand and debug.

a read **b** reading **c** to read

2 I went on the course _____ how to be a better programmer.

a learn **b** to learn **c** for to learn

3 I'm not interested in _____ that computer language.

a learn **b** learning **c** to learn

4 He refuses _____ the project with me.

a do **b** doing **c** to do

5 The engineers warned the employees not _____ the cables.

a touch **b** touching **c** to touch

6 They may not _____ to the conference.

a come **b** coming **c** to come

7 _____ in the IT industry is usually well paid.

a work **b** working **c** to work

8 Spyware can make your PC _____ more slowly.

a perform **b** performing **c** to perform

9 This program is too slow _____ the simulation.

a do **b** to do **c** for doing

10 Some people are desperate about _____ money more than learning how to program.

a making **b** to make **c** make

SPEAKING. Work in pairs. Student A reads about Python and [Student B](#) (at the end of the document) reads about Javascript. Do not read your partner's text. Then ask and answer questions to complete the chart.

Student A

Python was created by Guido van Rossum and first released in 1991. The name Python is inspired by the British comedy group Monty Python. Python is known for its clear syntax and readability, often described as "executable pseudocode." While not inherently "visual" in the same way as Visual Basic, Python offers powerful libraries like Tkinter, PyQt, and Kivy for creating graphical user interfaces. Instead of a drag-and-drop interface, developers use code to define and arrange interface elements such as buttons, labels, and windows. Python's versatility enables programmers to build a wide range of applications, from web applications and data analysis tools to desktop applications and games.

	Student A	Student B
Who developed the programming language and in what year was it first released?		
What does the name of the		

language (or a part of it) stand for or what inspired it?		
How is the language used to create user interfaces, and what are some examples of interface elements that can be created?		
What types of applications can programmers create using this language?		

Working in the IT industry.

READING. Read the memo. What is Lila writing about?

FROM: Chief Information Officer

TO: All Staff

DATE: June, 17th

SUBJECT: XBM Digital rules

Dear all,

As you know, many new staff joined us at XBM Digital, so now is a good time for a reminder about some of our rules.

Firstly, passwords are important to keep our network secure. Natasha is the person who sets the passwords and usernames at first. As soon as she gives them to you, you **need to** change the passwords which **have to** be alphanumeric and at least eight characters long. Also, you **should not** share them with anyone. You **might** find it difficult, but you **can** ask Natasha for help.

Secondly, the hardware that we have in our offices is delicate. You **mustn't** eat or put drinks near the computers. You **can** take short breaks for eating and drinking. You **must** do it in the dining room where all our staff eat everyday . You **don't have to** spend time cleaning your stuff, Rudy and Mike **will** take care of that for you.

Best regards,

Lila Davis

GRAMMAR. Modal verbs

FORM. Look at the words in **bold** and complete the rule.

- The verb that follows a modal verb is...
 - a) Present participle

- b) Past participle
c) Infinitive verb

FUNCTION. Complete the chart with examples from the text and analyse the function of each modal verb.

<i>Example</i>	<i>Function</i>
you (1) _____ it in the dining room... you (2) _____ the passwords	Personal obligation / strong necessity
...which (3) _____ alphanumeric...	External obligation
You (4) _____ or put drinks near the computers.	Prohibition
You (5) _____ time cleaning your stuff	Lack of necessity / Lack of obligation
you (6) _____ them with anyone. You should buy a new computer.	Advice
You (7) _____ short breaks for eating and drinking. I could not swim when I was a child. (Past) You will be able to program in nearly every language when you finish this course. (future)	Ability / inability
You (8) _____ it difficult I may go to the party, but I still don't know.	Possibility
Can I go to the toilet, please? Could you tell me where the restroom is? May I help you?	Request / Asking for permission / ask for or offer help

LANGUAGE WORK.

a) Complete the sentences with the appropriate modal verbs. There's an extra modal verb that you don't need to use:

Should - can - don't have to - might - have to - must - mustn't - doesn't have to

- 1) We _____ have a picnic tomorrow, but it depends on the weather. (possibility)
- 2) You _____ wear a suit to the interview. (Advice)
- 3) Mary! you _____ come to your aunt's birthday next Saturday night. (Lack of necessity)

- 4) You _____ smoke here! (Prohibition)
- 5) _____ I go to the toilet, please? (request/ask for permission)
- 6) Lisa _____ tidy her room before she goes out! (personal obligation)
- 7) In Britain you _____ drive on the left. (external obligation)

b) Write sentences...

a) ...giving **advice** to a friend as regards the clothes to wear on a party.

b) ...**asking for permission** to do something in a **polite** way.

c) ...describing a certain **ability** of your friend.

d) ...a rule that you obey at work/university. (**external obligation**)

e) ...something that you do because of **personal obligation**.

f) ...something that is **prohibited** to you.

g) ...something that you don't need to do. (**lack of necessity**)

GRAMMAR. Relative clauses

What is the difference between a sentence and a clause? Now, look at the underlined words in the memo above and choose the correct options to complete the rules.

1. The words **that, where, who** and **which**...
 - a) Introduce a new clause with information about something mentioned previously.
 - b) Introduce a new clause with information about something we didn't mention.
2. The relative clauses are found...
 - a) At the beginning of a sentence.
 - b) Following a previous clause or noun phrase.
3. The relative clause can be...
 - a) part of the subject.
 - b) part of the predicate.
 - c) both previous options.
4. We use the relative pronoun **who** (that can usually be replaced by **that**)...
 - a) To refer to things.
 - b) To refer to people.
5. We use the relative pronoun **which** (that can usually be replaced by **that**)...
 - a) To refer to things.
 - b) To refer to people.
6. We use the relative pronoun **where** (that **can't** be replaced by **that**)...
 - a) as the subject of the relative clause

b) followed by a subject in the relative clause.

Defining and non-defining relative clauses

- We use **defining relative clauses** to give essential information about someone or something. It usually comes immediately after the noun it describes. We don't use commas to mark its beginning or end.

*She is the woman **who** I met with yesterday.*

- We often leave out the relative pronoun when it is the object of the verb:

They're the people (that/who) she met at Jon's office.

Here are some bugs (that/which) the developer has identified.

- We use **non-defining relative clauses** to give extra information about the person or thing. It is not necessary information. We don't need it to understand who or what is being referred to. We use commas to mark the beginning and end of the non-defining relative clause. We can't use the relative pronoun 'that' to introduce a non-defining relative clause.

Clare, who I work with, is in charge of the project.

LANGUAGE WORK. Relative clauses

a) Complete the sentences with *who*, *which* or *where*.

- 1) Do you know the man _____ lives next door?
- 2) That's the gallery _____ had the Leonardo Da Vinci exhibition.
- 3) Are those the people _____ are selling their house?
- 4) Do you know a good restaurant _____ is open on Saturday night?
- 5) Is that the bus _____ goes to the airport?
- 6) We walked past the school _____ their children go.
- 7) She's the woman _____ everyone is talking about.
- 8) I took my laptop back to the shop _____ I bought it.
- 9) Antwerp is the city _____ I lived as a child.
- 10) Is there someone _____ can speak Arabic in your class?

b) Write sentences with **defining relative clauses** using the prompts given:

a) Bus = central cable / link / computers

____ A bus is a central cable which/that links computers _____

b) Customers = people / buy / things

c) Production facilities = places / manufacture / products

d) WAN = type of network / cover / large areas

HOMEWORK TO HAND IN

WRITING. Describing functions and features.

Read the instructions below and write a description of the Sony PlayStation 3 controller.

Describing functions and features

We can describe functions and features by using:

- a preposition + a gerund
 This is a device **for controlling** the cursor and selecting items on the screen.
- An infinitive to express purpose
 It is used **to control**...
- A relative clause
 This device, **which is an input device**, is used to...
 This is a device **which** is used to...
- A modal verb + bare infinitive
 You **can connect** it to the computer...

We can also describe features like this:

- An optical mouse **has** an optical sensor instead of a ball underneath.
- It usually **features** two buttons and a wheel.
- A wireless mouse **operates/works** without cables.
- It **allows** the user **to** answer multiple-choice questions...



Sony PlayStation 3 controller

Functions

- control video games
- hold it with both hands, use thumbs to handle directional sticks and face buttons

Features

- six-axis sensing system (capable of sensing motion in six directions: up, down, left, right, forwards and backwards)
- wireless controller (Bluetooth)
- USB mini port and cable for wired play and automatic battery charging

Unit N° 2 - Gaming

SPEAKING. The following pictures were extracted from different popular video games of all times. In pairs or small groups, decide which names they belong to. Then, guess their years of release.



Discuss with your partners:

- Have you ever played any of the games above? If so, which ones? If not, which ones would you like to try?
- Which one do you think is the best? Why?

READING COMPREHENSION. a) Read the following article about game genres. Which genres do the games from the previous ex. belong to?

Game Genres

With the rapid evolution of the gaming industry, genres have been expanding and blending, making it increasingly difficult to classify every game into a single category. In this article, we'll explore the major genres that define today's gaming landscape.

First-person shooters (FPS) and **Action games** have been dominating the market. Iconic franchises like *Call of Duty* and *Battlefield* remain strong, while newer hits like *Apex Legends* and *Valorant* have been

redefining competitive FPS gameplay. In the Action genre, open-world experiences such as the *Grand Theft Auto series*, *Red Dead Redemption 2*, and the highly anticipated *GTA V* have been pushing boundaries in storytelling and realism.

Role-playing games (RPGs) have remained a cornerstone of both console and PC gaming. Titles like *Elden Ring*, *Cyberpunk 2077*, and *Baldur's Gate 3* have set new standards for immersive storytelling and player choice. The rise of live-service RPGs, such as *Genshin Impact*, has revolutionized the genre, offering expansive worlds and frequent content updates. **Massively multiplayer online RPGs (MMORPGs)** have been continuing to thrive, with games like *Final Fantasy XIV* and *World of Warcraft* maintaining dedicated player bases.

Adventure and **Puzzle games** have been experiencing a resurgence, thanks to indie developers and innovative mechanics. Story-driven experiences like *The Legend of Zelda: Tears of the Kingdom* and puzzle-based narratives like *The Witness* and *Portal* have been captivating players. The party game genre has also been evolving, with titles like *Among Us* and *Fall Guys* introducing new social and competitive elements.

Sports games remain a significant part of the industry, with franchises like *FIFA* (now EA Sports FC), *NBA 2K*, and *Madden NFL* leading the way. Meanwhile, racing games such as *Forza Horizon* and *Gran Turismo* have been pushing the limits of realism and gameplay depth.

The **Simulation** genre has been expanding beyond traditional life simulators. While *The Sims 4* remains a dominant force, newer titles like *Stardew Valley* and *Animal Crossing: New Horizons* have been offering fresh takes on virtual life. Flight simulators and city-building games, like *Microsoft Flight Simulator* and *Cities: Skylines*, have also been surging in popularity.

Strategy games, once primarily a PC domain, have been gaining traction on consoles thanks to improved controls and cross-platform play. Games like *Civilization VI*, *XCOM 2*, and *Total War: Warhammer III* have been attracting tactical minds, while real-time strategy (RTS) games like *Age of Empires IV* have been maintaining the genre's legacy.

Fighting games have entered a new era with advanced mechanics and competitive esports scenes. Titles like *Street Fighter 6*, *Tekken 8*, and *Super Smash Bros. Ultimate* have been pushing the genre forward, incorporating refined gameplay and online matchmaking for global tournaments.

As gaming technology advances, the landscape of genres will continue to evolve. By 2035, developers will have fully integrated artificial intelligence, virtual reality, and cloud-based gaming into mainstream titles. We will have witnessed the rise of entirely AI-generated open-world games, where player choices shape narratives in unprecedented ways. Furthermore, the boundaries between genres will have blurred even more, creating experiences that defy traditional classification, ensuring that gaming remains one of the most dynamic forms of entertainment.

b) These statements about games are all mistaken. Correct them.

1) Roleplay games are currently the most popular ones.

2) Adventure and Puzzle games have always remained as the dominant choice.

3) Forza Horizon is a Strategy game.

4) The Sims Series is the least popular in the Simulation category.

5) Strategy games can only be played using a PC.

6) Street Fighter 6 is not a fighting game.

GRAMMAR. Present Perfect Continuous

Reflect upon grammar. Read the text above again and find examples of sentences which contain Present Perfect Simple tense.

- Do you remember its structure?
- When do you use Present Perfect Simple?
- Can you identify any other similar structure in the text? How are they different in terms of meaning?

FORM. Complete the chart with examples from the text.

Affirmative	Negative	Interrogative	Short answers
Adventure and Puzzle games (1) _____ _____ a resurgence, thanks to indie developers and innovative mechanics. (I/we/you/they)	First-person shooters (FPS) and Action games have not (haven't) been dominating the market.	Have FPS and Action games been dominating the market?	Yes, they have . No, they haven't .
The Simulation genre (2) _____ _____ beyond traditional life simulators. (he/she/it)	The party game genre has not (hasn't) been evolving .	Has it been evolving?	Yes, they has . No, they hasn't .

FUNCTION. Choose the rules that apply to Present Perfect Continuous tense.

- a) To talk about recent activities when the effects can still be seen.
- b) To talk about single actions that started and finished in a particular moment in the past.
- c) With action verbs to emphasise the completion of an event in the recent past.
- d) To emphasise how long an action has been going on or if it has been repeated many times.
- e) To suggest that an activity is temporary.
- f) To talk about an unfinished period of time up to the present, but not necessarily to suggest that the action will continue for a short period of time.
- g) To suggest that an action is not complete while it should have been completed by the time of speaking.

Which of the previous rules could explain Marge's question?

LANGUAGE WORK: the present perfect

- a) Choose the correct words in brackets to complete these sentences.

- 1 He ('s never liked /'s never been liking) Maths.
- 2 They ('ve worked/'ve been working) on the project all day.
- 3 John ('s used/'s been using) the computer for hours - he looks really tired.
- 4 How many emails (have you written/ have you been writing) today?
- 5 She ('s written/'s been writing) this essay since 9 o'clock.
- 6 They ('ve interviewed/'ve been interviewing) five candidates today.



- b) Make questions using these prompts.

1 ever / live or work in another country?

 2 ever / have a bad job interview?

 3 ever / do a job you hated?

 4 how long / study English?

 5 how long / study programming?

 6 how many emails / receive today?

 7 how many jobs / apply for this year?

SPEAKING. Work with a partner. Ask and answer the questions in ex b

GRAMMAR. Future perfect

Look back to the last paragraph of the text “Game genres”. Can you name the advances that will occur in gaming technology?

Reflect upon grammar. Read this sentence extracted from the text and choose the best answer to the following question:

“By 2035, developers will have fully integrated artificial intelligence, virtual reality, and cloud-based gaming into mainstream titles.”

- What will happen first?
 - a) 2035 arrives
 - b) developers fully integrate AI, virtual reality and cloud-based gaming into mainstream titles.

FORM.

Affirmative	Negative	Interrogative	Short Answers
We will have witnessed the rise of entirely AI-generated open-world games.	We will not (won't) have witnessed the rise of entirely AI-generated open-world games.	Will we have witnessed the rise of entirely AI-generated open-world games?	Yes, we will . No, we won't .

FUNCTION. We use the future perfect form when we look back to the past from a point in the future. We usually use a time phrase, for example:

- **by** tomorrow, 2035, 7 o'clock, this evening...
- **before** Tuesday, July, 2030...
- **this time** next week, year, Sunday...
- **in** ten years (time), two weeks (time)...

LANGUAGE WORK. a) Choose the event which happens first.

- 1) By the time the new gaming console is released, developers will have created hundreds of new games.
 - a) The new gaming console is released.
 - b) Developers create hundreds of new games.

- 2) By the end of the tournament, the champion will have defeated all opponents.
 - a) The champion defeats all opponents.
 - b) The tournament ends.

- 3) You will have unlocked all the special abilities by the time you reach level 50.
 - a) You unlock all the special abilities.
 - b) You reach level 50.

- 4) By midnight, we will have completed the final mission in the game.
 - a) We complete the final mission.
 - b) It is midnight.

b) Complete the sentences using the correct form of the verbs in brackets.

- 1- By the end of the year, I (save) _____ £500.
- 2- By this time next year, my father (spend) _____ five years in France.
- 3- Scientists (possibly/develop) _____ a new vaccine for AIDS in twenty years time.
- 4- By the end of this week, I (not be able) _____ to finish the report.
- 5- When you get to the airport, _____ Paul (already/arrive) _____?
- 6- I (leave) _____ by seven.

LISTENING. Call centres: are you talking to an AI?

Have you ever phoned up a company and had your call held in a queue? Could AI make this less painful?

VOCABULARY. Match the words (1-6) to its meanings (a-f).

- 1) yelling

- 2) pacify (someone)
- 3) in the near future
- 4) as a matter of fact
- 5) empathetic
- 6) you know

- a) (filler word) used to give the speaker time to think, or to express uncertainty
- b) shouting
- c) used to add emphasis to a statement, to give more detail about what has just been said, or to introduce something that contrasts with it.
- d) calm down someone who is angry.
- e) able to put yourself into someone else's position and share their feelings or experiences.
- f) very soon, within a short time

Approximately how many people work in call centres globally? Answer the question. Then, listen and check your answers.

- a) 7 million?
- b) 17 million? or,
- c) 27 million?

LISTENING AND SPEAKING HOMEWORK. Throughout Inglés I and Inglés II you've been learning a lot of vocabulary, grammar and skills in general that can place you between a pre-intermediate and an intermediate learner of English. It's time to enhance your listening and speaking skills in order to become an independent learner. From now on, every class you'll start teaching the whole class a new expression that is currently being used. Go to https://www.bbc.co.uk/learningenglish/features/the-english-we-speak_2025, choose a video, listen to it as many times as you need and the following class you'll teach your classmates the expression you've learnt.

HOW TO SAY IT

The video I chose was about...
The expression I've learnt is...
It means that...
An example of "..." is...
It is similar to...
It is the opposite to...

Unit N° 3 - A little bit of history

READING COMPREHENSION. Read the text. Did you know any of the information provided?

A little bit of History*

The Facultad Regional de Avellaneda was created on March 31, 1955 by Resolution 382/55 of the National Commission for Learning and Professional Orientation, under the jurisdiction of the Universidad Obrera Nacional. The establishment of this college was part of one of the objectives of the Second Five-Year Plan of Juan Domingo Perón's government, which aimed at the creation of institutes like this one to provide graduates with advanced knowledge that would qualify them to lead and manage activities within the national economic framework.

However, to fully understand its origins, we must go back a few years. On August 19, 1948, the National Workers' University had been created through the enactment of Law 13,329. This law designated it as an institution of higher education "...designed to provide the solution the country was seeking: new professionals in the field of engineering." This was a particularly significant period in Argentina's history, marked by a flourishing and expanding industry, where the import substitution model was at its peak.

"The government's policy was clear: on the one hand, to create living conditions far superior to those of the time, producing a true social revolution, and on the other, to provide specialized labor that the emerging national industry required. A class of workers capable of deepening the country's development, destined to meet an increasingly important demand."

This new social context "...led a group of professionals, teachers from the Otto Krause Industrial School, to unite in search of a high-level solution." Thus, the objective was to:

- Provide a higher education path for graduates of industrial schools or the final cycle of institutions under the National Commission for Learning and Professional Orientation, particularly in engineering and its various specializations.
- Offer access to university education to a large number of students, mostly from lower-income and middle-class backgrounds.

The creation of the Universidad Obrera Nacional brought these goals to fruition.

However, with the rise of the 1955 dictatorship, this university—born within a clear national political project—came under heavy scrutiny and was on the verge of disappearing. This abrupt shift in national policy led to an institutional crisis within the Universidad Obrera Nacional and, consequently, the newly established Facultad Regional de Avellaneda.

To address this crisis, Law 14,855, enacted on October 14, 1959, granted the university a new autonomous legal framework, aligning it with other national universities. As a result, it ceased to depend on the former National Commission for Learning and Professional Orientation and was renamed the National Technological University (Universidad Tecnológica Nacional, UTN).

The city of Avellaneda was chosen as the site for one of the first colleges of the National Technological University (UTN). For a long time, it was the only national university-level institution in this area of Gran Buenos Aires. Located in the heart of the city for more than fifty years, it now has two buildings.

The main building, located in front of the plaza and shared with other educational institutions, hosted its first courses on May 5, 1955, marking the origins of the college known as the "Workers' University".

Due to a rapid increase in student enrollment, this building soon became insufficient. As a result, a new campus was built in Villa Domínico, at the intersection of Mitre and Ramón Franco Avenues, on land granted to the university for the college's operation.

* Extracted and Translated from <https://fra.utn.edu.ar/institucional/>

b) Read the text again and answer the following questions:

1) When and why was the Facultad Regional de Avellaneda created?

.....

2) What was the main objective of the Second Five-Year Plan regarding higher education?

.....

3) What was the purpose of the Universidad Obrera Nacional, and when was it established?

.....

4) How did the government's industrial and social policies influence the creation of this university?

.....

5) What challenges did the Universidad Obrera Nacional face after the 1955 dictatorship?

.....

6) How did Law 14,855 change the legal status of the university?

.....

7) Why was Avellaneda chosen as the location for one of the first colleges of UTN?

.....

8) What led to the construction of a second campus in Villa Domínico?

.....

GRAMMAR. The passive

Reflect upon grammar. Look back at the first sentence in the text. Can you identify its verb phrase? Which is the first verb? In which form is the second verb? In what tense is the sentence written? How do you know?

The object of an active sentence becomes the subject of the passive sentence. We can use the word **by** to say who or what did something in a passive sentence.

SIMPLE PRESENT ACTIVE: *"I write a sentence in the simple past."* (S + V + O)

SIMPLE PRESENT PASSIVE: *"The sentence is written in the simple past (by me)."* (old O + IS/ARE/AM + PAST PARTICIPLE + by + old S)

SIMPLE PAST ACTIVE: *"Two students created Google in 1998."* (S + V + O)

SIMPLE PAST PASSIVE: *"Google was created by two students in 1998."* (old O + WAS/WERE + PAST PARTICIPLE + by + old S)

FUTURE SIMPLE ACTIVE: *"In 2026, Elon Musk will send the first orbital flight of Starship to Mars."* (S + V + O)

FUTURE SIMPLE PASSIVE: *"In 2026, the first orbital flight of Starship will be sent to Mars."* (old O + WILL BE + PAST PARTICIPLE + by + old S)

LANGUAGE WORK.

A) Are these sentences Active or Passive?

- 1) Steven likes to play basketball. ____
- 2) Bingo is played in Britain. ____
- 3) He lost his keys yesterday. ____
- 4) A house will be bought by them. ____
- 5) A letter was written. ____
- 6) These cars are produced in Italy. ____
- 7) Lots of houses were destroyed. ____
- 8) I will go to Egypt one day. ____

B) Rewrite the Active sentences into Passive form:

1)- They understand Spanish.

2)- My friend bought a new car.

3)- John cleaned the bathroom.

4)- The teacher closes the window.

5)- The professor will show the students an old bone.

6)- Snow covered the mountains.

7)- They will send astronauts to the moon.

8)- Frida Kahlo painted around 200 masterpieces.

GRAMMAR. Past Perfect

Reflect upon grammar. Look at the following sentence. Which of the two actions happened first? In which verb tense is the second action written?

“The government **had created** the Universidad Obrera Nacional before the Facultad de Avellaneda **was established.**”

FORM

Affirmative	Negative	Interrogative	Short Answer
By the time the dictatorship began in 1955, the university had already started offering courses.	By the time the dictatorship began in 1955, the university had not (hadn't) already started offering courses.	By the time the dictatorship began in 1955, had the university already started offering courses?	Yes, it had . No, it hadn't .

FUNCTION. Choose the correct option. We use the past perfect to talk about...

- Something that happened before a specific time in the past.
- Something that happened a very long time ago.

LANGUAGE WORK. Complete the sentences with the correct past simple or past perfect of the verbs in brackets.

- When we _____ (arrive), our parents _____ already _____ (eat).
- I _____ already _____ (have) two jobs by the time I _____ (be) 18.
- I _____ just _____ (finish) my work when you _____ (phone).
- By the age of five, my brother _____ already _____ (learn) how to use a computer.
- We _____ (not finish) our exam when the bell _____ (ring).
- I _____ (go) to New York last summer. I _____ never _____ (go) there before.
- Why _____ we _____ (not see) this movie before last night?
- _____ you already _____ (meet) Sally when you _____ (start) university?

SPEAKING. Talking about memories. Complete the sentences about your experiences. Then write suitable questions to ask your partner about their experiences.

- By the time I was ..., I had already...
- When I ... for the first time, I had never ...
- When ..., I hadn't ... yet.
- When..., I had just ...

Unit N° 4 - Networks

LISTENING. Listen to two technicians talking about video conferencing technology. Choose the correct option. (English for IT 2 - track 32)

- 1) At the beginning of the conversation, the woman mentions some types of video conferencing systems. They are...
 - a) Dedicated system and desktop system.
 - b) Dedicated system, desktop system and compression system.
 - c) Dedicated system only.
- 2) To use a desktop system you need...
 - a) A special room with its own hardware.
 - b) An ordinary PC and some hardware.
 - c) A set of high definition monitors and a video camera for each participant with a remote control.
- 3) MCU (Multipoint control unit) will be useful if...
 - a) You need to hold a video conference between only three different locations.
 - b) You need to hold a video conference between two different locations.
 - c) You need to hold a video conference between more than three locations.
- 4) The man says that if they had a video conferencing system, ...
 - a) They would use less bandwidth.
 - b) They could connect to their Tokio, Dubai and Paris offices.
 - c) They might see each other's faces.

FORM

Type 0: "If you heat ice, it melts."

If + _____ , _____

Type 1: "If you don't hurry, you will miss the train."

If + _____ , will/can/may + _____

Type 2: "If I had a lot of money, I would travel all over the world."

If + _____ , would/could/might + _____

Type 3: "If I had known you had a swimming pool, I would have brought my bikini."

If + _____, would/could/might + _____ + _____

FUNCTION. True or false? Justify the false ones.

1) Conditional sentences always have only one clause.

2) Conditional sentences describe the result of something that might happen or might have happened but didn't.

3) Conditionals type 0 are used to describe past actions.

4) Conditional sentences type 1 describe a possible condition and its probable result.

5) Conditionals type 2 describe a real situation in the past.

6) Conditional sentences type 3 refer to an unreal past condition and its probable result in the present.

LANGUAGE WORK. Complete the conditional sentences with the correct form of the verbs in brackets:

- a) If you _____ (mix) yellow and red, you _____ (get) orange. (Type 0)
- b) If I _____ (know) It was going to rain, I _____ (bring) my umbrella. (Type 3)
- c) Simon _____ (go) to Mendoza this weekend if he _____ (find) a cheap flight. (Type 1)
- d) If you _____ (not help) endangered species, they _____ (disappear) soon. (Type 1)
- e) I _____ (study) harder if I _____ (have) more time. (Type 3)
- f) My sister _____ (have) a baby if she _____ (work) less hours. (Type 2)
- g) The water _____ (boil) if you _____ (heat) it up to 100°C. (Type 0)
- h) If I _____ (be) you, I _____ (rent) a new apartment. (Type 2)

LISTENING. A) Listen to an extract from a lecture about networks and answer these questions.
 (Infotech – Track 34)

1) What does LAN stand for?

_____.

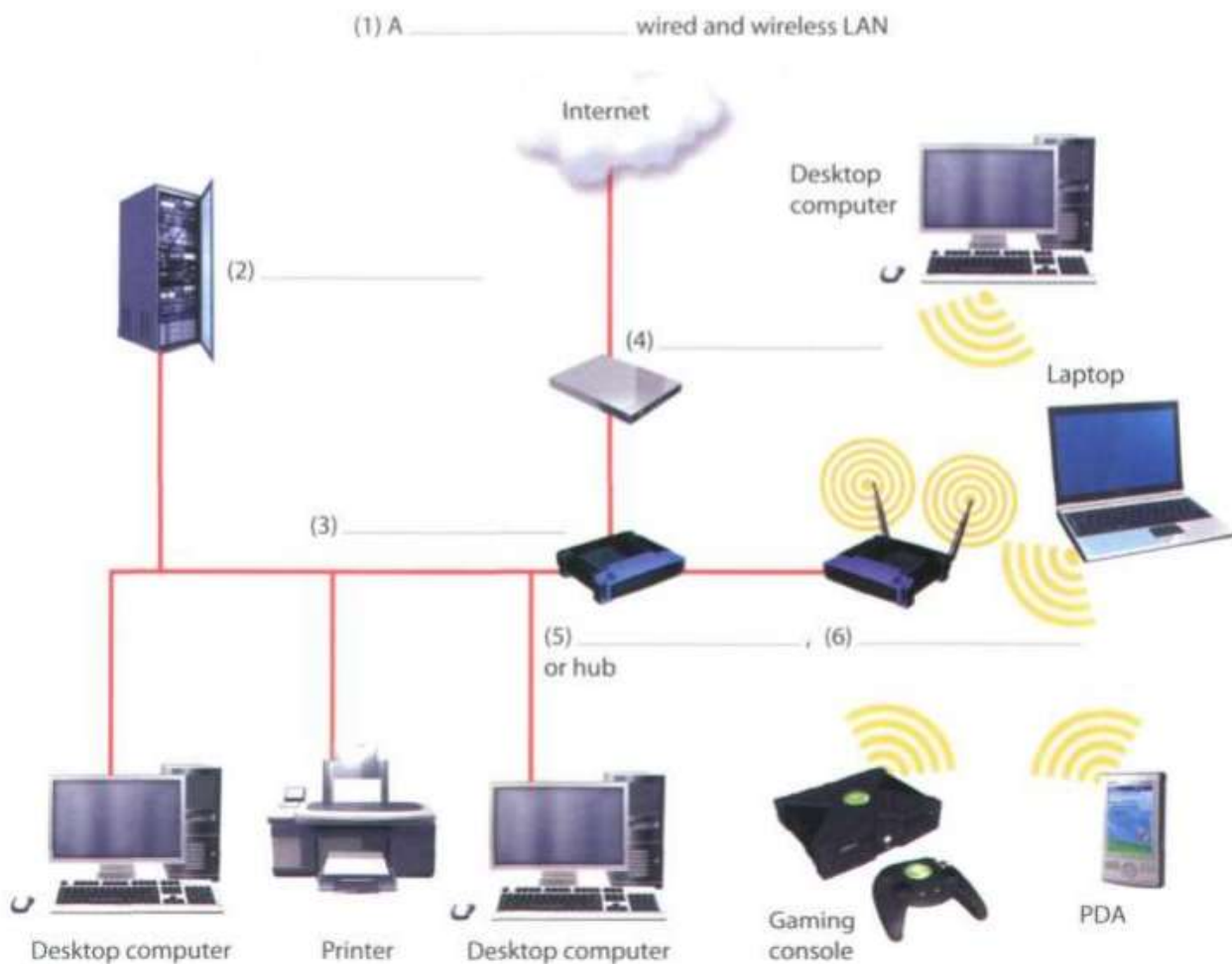
2) Where are LANs usually located?

_____.

3) What is the difference between a wired LAN and a wireless LAN?

B) Listen again and label the elements of this LAN

- 1) -----
- 2) -----
- 3) -----
- 4) -----
- 5) -----
- 6) -----



READING. Read the following text. Why can it cause a slow connection?

Setting up a network requires careful planning and the right equipment. First, you need to **set up** a connection by choosing a reliable internet provider. Then, you must **plug into** the main router to access the network. Many companies **carry out** security checks to ensure that their networks are safe.

When creating an online account, users must **sign up** by providing their details and **fill in** registration forms. After that, they need to **type in** their username and password to log in. If they want to test the network speed, they can **try out** different online tools.

If there is a problem, IT specialists **look for** solutions and **find out** what is causing the issue. Sometimes, a slow connection is due to multiple devices taking up too much bandwidth. To fix this, users should **switch off** unnecessary devices and **turn on** only the essential ones.

A network is **made up** of various components, including routers, servers, and connected devices. Whether at work or home, staying connected is essential in today's digital world.

VOCABULARY. Read the text again and match the phrasal verbs below (1-12) to their meanings (a-l). Try to infer meaning from context.

- | | |
|----------------|-------------------------------------|
| 1) set up | a) connect |
| 2) plug into | b) start or use experimentally |
| 3) carry out | c) establish |
| 4) sign up | d) write the necessary information |
| 5) fill in | e) register, enroll in a service |
| 6) type in | f) execute |
| 7) try out | g) stop the operation of something |
| 8) look for | h) learn, discover |
| 9) find out | i) constitute, form |
| 10) switch off | j) start the operation of something |
| 11) turn on | k) occupy |
| 12) make up | l) search |

Phrasal verbs

A phrasal verb is an idiomatic phrase consisting of a verb and another element, typically either an adverb or a preposition.

- The meaning of some verbs with particle (often called phrasal verbs) can be easily understood from its two parts.

***Look at** the photos.*

*A network **consists of** two or more...*

*Separate networks are **linked over** a public network, the Internet.*

- However, many phrasal verbs have an idiomatic meaning, not predictable from the meaning of its parts.

carry (= transport); **carry out** (= execute)

*Computers **carry out** the programs...*

- Certain particles have similar meanings, regardless of the verb (**on/off, in/out**, etc.).

turn on / switch on, turn off / switch off

- When the verb has a *preposition* associated with it, the preposition must precede the object:

*You can **look for** information on the Web. (not **look** information **for**)*

*Hackers might **break into** your PC (not **break** your PC **into**)*

- When the particle is an *adverb*, it can precede or follow the direct object:

*You need to **type in** your username/ ... **type** your username **in**.*

*You can **look up** words in a dictionary/ ... **look** words **up** in a dictionary.*

***Turn on** the computer./ **Turn** the computer **on**.*

- If the direct object is a pronoun, the particle must follow it

*You need to **type it in**. (not: **type in it**)*

HOMEWORK. Click on this link <https://www.liveworksheets.com/w/en/english-second-language-esl/1430449> and do the worksheet. Here you'll revise present perfect simple, present perfect continuous, simple past and past perfect. You'll also learn some more phrasal verbs. ***The worksheet should be completed using capital letters and do not use contractions.***

Unit N° 5 - The evolution of Information Technology

READING COMPREHENSION. In groups, read the text and answer the following questions:

The Evolution of Information Technology

In today's interconnected world, information technology (IT) plays a pivotal role, **so** the way we communicate, work, and live is still being transformed. Throughout history, remarkable advancements have been experienced by IT, driven by a combination of technological breakthroughs and societal demands. In this text, the evolution of IT will be explored, highlighting key developments **and** their impact on various aspects of our lives. **Although** technology has helped us a lot, privacy and security don't have to be underestimated.

To begin with, the origins of IT can be traced back to the mid-20th century when computers emerged as powerful tools for data processing. The first computers, **which** were the size of entire rooms, were mainly used for complex calculations and data storage. **After that**, in the 1970s, personal computers (PCs) revolutionized the IT landscape **because** they brought computing power to individuals. This

marked the beginning of a new era, **where** PCs became increasingly affordable, compact, and user-friendly.

Over the years, IT has witnessed significant advancements which were facilitated by rapid technological innovation. One notable breakthrough occurred in the 1990s with the advent of the World Wide Web. This revolutionary platform enabled seamless information sharing and transformed the way we access knowledge, connect with others, and conduct business. **As a result**, a vast array of online services and applications emerged, shaping the digital landscape we navigate today.

In recent years, IT has embraced groundbreaking technologies **such as** artificial intelligence (AI), the Internet of Things (IoT), and cloud computing. AI, **for instance**, has the potential to automate repetitive tasks, enhance decision-making processes, and even simulate human-like intelligence. The IoT, **on the other hand**, connects various devices and sensors, enabling efficient data collection, analysis, and communication. **Additionally**, cloud computing offers a scalable and flexible infrastructure for storing and accessing data and applications. It is important to note that along with the benefits that have been brought by these technologies; **however**, there are **also** concerns regarding privacy and security in the realm of IT.

In conclusion, IT has undergone a remarkable evolution, transforming our society in profound ways. From the emergence of computers to the digital age of the internet and the latest advancements in AI, IoT, and cloud computing, IT continues to shape our lives and drive innovation. **Therefore**, it is crucial to embrace and adapt to these technologies, harnessing their potential to create a more connected and efficient world.

- 1) What role did personal computers play in the evolution of IT?

- 2) When did the World Wide Web emerge and how did it transform information sharing?

- 3) What are some of the groundbreaking technologies that IT has embraced in recent years?

- 4) How does artificial intelligence (AI) contribute to the advancement of IT?

- 5) What are the benefits of cloud computing in the IT landscape?

GRAMMAR. Passive voice.

Look back at the text and underline all the verbs “to be” followed by a past participle. In which tenses are these sentences written?

FORM. Complete the chart using the correct form of the passive voice structure in each verb tense:

	Active sentence	Affirmative	Negative	Interrogative
Present continuous	1. The employee is sending the email. 2. The user is downloading the file from the internet.	1. The email is being sent by the employee. 2. The file _____ from the internet by the user.	1. The email isn't being sent by the employee. 2. The file _____ from the internet by the user.	1. Is the email being sent by the employee? 2. _____ the file _____ from the internet by the user?
Past Continuous	1. The employee was sending the email. 2. The user was installing the software on the computer.	1. The email was being sent by the employee. 2. _____	1. The email wasn't being sent by the employee. 2. _____	1. Was the email being sent by the employee? 2. _____
Present Perfect	1. The employee has sent the email. 2. The technicians have fixed the malfunctioning server.	1. The email has been sent by the employee. 2. _____	1. The email hasn't been sent by the employee. 2. _____	1. Has the email been sent by the employee? 2. _____
Past Perfect	1. The employee had sent the email. 2. They had saved the document to the cloud storage.	1. The email had been sent by the employee. 2. _____	1. The email hadn't been sent by the employee. 2. _____	1. Had the email been sent by the employee? 2. _____

Future perfect	1. The employee will have sent the email. 2. The user will have visited the website.	1. The email will have been sent by the employee. 2. _____ _____ _____ _____ _____	1. The email will not (won't) have been sent by the employee. 2. _____ _____ _____ _____ _____	1. Will the email have been sent by the employee? 2. _____ _____ _____ _____ _____
Modal verbs	1. The employee must send the email. 2. The programmer can write the code for the new software update.	1. The email must be sent by the employee. 2. _____ _____ _____ _____ _____	1. The email doesn't have to be sent by the employee. 2. _____ _____ _____ _____ _____	1. Must the email be sent by the employee? 2. _____ _____ _____ _____ _____

LANGUAGE WORK. Fill in the blanks with the correct passive form of the verbs in brackets.

1. **Present Continuous:**

- A new security system _____ (**install**) in the company's network right now.
- Important files _____ (**not transfer**) to the cloud at the moment.
- _____ the new software _____ (**test**) on all devices?

2. **Past Continuous:**

- A critical update _____ (**apply**) to the system when the power went out. (*affirmative*)
- The server _____ (**not monitor**) properly when the cyberattack occurred.
- _____ the Wi-Fi network _____ (**repair**) during the outage?

3. **Present Perfect Simple:**

- Several security measures _____ (**implement**) in the last few months.
- The password policy _____ (**not change**) since last year.
- _____ all the reports _____ (**upload**) to the database?

4. **Past Perfect Simple:**

- a) By the time the IT team arrived, the malware _____ (**remove**) from the system.
- b) The unauthorized access _____ (**not detect**) before the system crashed.
- c) _____ the backup files _____ (**save**) before the server shut down?

5. **Future Perfect:**

- a) By next year, a more advanced firewall _____ (**develop**) to improve security.
- b) Sensitive data _____ (**not encrypt**) by the end of the month if they don't act quickly.
- c) _____ all the necessary updates _____ (**complete**) before the new system launches?

6. **Modal Verbs:**

- a) Stronger passwords _____ (**should/use**) to prevent hacking attempts.
- b) Some network configurations _____ (**must/not/change**) without administrator approval.
- c) _____ remote access _____ (**can/disable**) to protect user data?

GRAMMAR. Connectors

CONNECTORS, also called conjunctions, are words or expressions which we use to establish different types of relationships between words or sentences. When we use connectors appropriately, we organise what we say in a more logical, understandable way.

Reflect upon punctuation. Look at the words in bold in the first text and then choose the correct options to complete the punctuation rules.

- 1) Use a **comma / semicolon** before words like *but* or *so*.
 - 2) Use a **full stop / comma** before connectors like *where*, *when*, *which*, *even though*, *since* and *although* when they come in the middle of a sentence.
 - 3) **Use / don't use** a comma after words like *however*, *therefore*, *additionally* and *after that* at the beginning of a sentence.
 - 4) Use a **comma / semicolon** before words like *however*, *that is* and *on the other hand* when they come in the middle of a sentence.
 - 5) When words like *since*, *although* and *because* appear at the beginning of a sentence, use a **comma / full stop** at the end of the clause, not after the word.
- Highlight all the connectors in the following examples and then complete the chart adding the connectors used in the text:

Type of connector	Examples
Contrast	1. _____ technology has helped us a lot, privacy and security don't have to be underestimated. 2. Though smartphones have revolutionized communication, they can also be a source of distraction. 3. In spite of the increasing reliance on digital communication, many people still value face-to-face interactions. 4. Despite the fact that cybersecurity measures have improved, online threats continue to pose a significant risk. 5. On the one hand, technology has made our lives more convenient, but on the other hand, it has also led to concerns about privacy. 6. Instead of relying solely on traditional advertising methods, many companies now invest heavily in online marketing. 7. In contrast to traditional paper-based processes, digitalization has streamlined administrative tasks and increased efficiency. 8. It is important to note that [...] technologies; _____, there are also concerns regarding privacy and security in the realm of IT.
Addition	1. Moreover, the Internet of Things (IoT) enables seamless connectivity between various devices, leading to improved efficiency and automation. 2. Furthermore, advancements in artificial intelligence (AI) have revolutionized industries such as healthcare, finance, and transportation. 3. In addition to the convenience of online shopping, e-commerce platforms provide a wide range of choices and competitive pricing. 4. Besides its role in communication, social media has also become a platform for information sharing and online communities. 5. [...] highlighting key developments _____ their impact on various aspects of our lives. 6. There are _____ concerns regarding privacy and security in the realm of IT. 7. _____, cloud computing offers a scalable and flexible infrastructure for storing and accessing data and applications.
Example	1. AI, _____, has the potential to automate repetitive tasks, enhance decision-making processes, and even simulate human-like intelligence. 2. IT has embraced groundbreaking technologies _____ artificial intelligence (AI), the Internet of Things (IoT), and cloud computing. 3. Cloud computing can be defined as the practice of using remote servers to store, manage, and process data. In other words, this provides on-demand access to resources and services. 4. Artificial intelligence (AI) involves the development of intelligent machines. That is to say, they are capable of performing tasks that typically require human intelligence, such as problem-solving and decision-making.

Sequence / Time	1. First of all / First(ly), let's discuss the basic principles of programming before diving into more complex topics. 2. Second(ly), once you have grasped the theoretical concepts, it's time to apply them through practical exercises. 3. Third(ly), reviewing and practicing regularly will help reinforce your understanding and improve your skills. 4. Later in the course, we will explore advanced topics that build upon the foundation we have established. 5. Afterwards / Then / Next, we will move on to discussing algorithms and their role in solving computational problems. 6. _____, in the 1970s, personal computers (PCs) revolutionized the IT landscape 7. Finally, we will conclude the course by working on a capstone project that applies the knowledge and skills gained throughout the semester. 8. Meanwhile / In the meantime, as you work on your assignments, don't hesitate to reach out to your classmates or instructor for assistance. 9. Before starting a coding project, it is important to plan and outline the steps involved. 10. While studying computer science, it is crucial to stay updated with the latest technological advancements. 11. Until you grasp the concept of recursion, solving certain programming problems may seem challenging.
Purpose	1. I am studying diligently so that I can improve my English skills. 2. She arrived early so as to secure a good seat for the test. 3. In order to succeed in the competitive job market, it is essential to acquire relevant skills and qualifications. 4. He practiced regularly in order to become a skilled developer.
Cause and effect	1. IT plays a pivotal role, _____ the way we communicate, work, and live is still being transformed. 2. The software had multiple bugs, and that is why it crashed frequently. 3. As a consequence of the coding error, the application failed to function properly. 4. He implemented a new algorithm, thus improving the performance of the system. 5. PCs revolutionized the IT landscape _____ they brought computing power to individuals. 6. The project was delayed because of a change in client specifications. 7. Owing to his expertise in programming languages, he developed a robust and efficient software solution. 8. On account of his meticulous testing, the software had minimal defects upon release.

LANGUAGE WORK: Read the following text and choose the correct connector in each case.

Understanding Algorithms: Loops, Counters, and Accumulators

An *algorithm* is a set of instructions designed to solve a problem. **(1) On the one hand / On the other hand**, algorithms must be efficient and structured properly to achieve the desired results. **(2) However / Moreover**, writing a correct algorithm requires understanding different programming concepts.

Loops are an essential part of algorithms. They allow a program to repeat a sequence of instructions **(3) although / because** they reduce redundancy and improve efficiency. **(4) Moreover / Therefore**, they are widely used in programming. **(5) For instance / That is to say**, a *for loop* can iterate through a list of numbers, performing calculations automatically.

Within loops, *counters* and *accumulators* play a key role. A *counter* keeps track of the number of iterations, **(6) because / thus** it is often used to control loop execution. **(7) Meanwhile / Because**, an *accumulator* stores a running total of values, such as adding all elements in a list.

Understanding how to use these elements correctly is crucial. A poorly designed loop may run indefinitely **(8) until / besides** a specific condition is met. **(9) In addition / Because**, using the wrong variable for accumulation can lead to incorrect results.

Mastering these concepts helps programmers develop efficient algorithms, **(10) that is to say / for instance**, it helps them optimize calculations and minimize execution time.

Unit N° 6 - Negotiating

A) Put the words in order to write complete sentences and questions an Interviewer may ask:

1) in five years / see / How / you / yourself / do / ?

2) been / Have / in charge of / ever / you / a team / ?

3) some questions / to ask / yourself. / I / would like / you/ that / Is / about / OK / ?

4) morning, / meet / Good / you. / to/ nice

5) work experience / tell / about / previous / your / Could / me / you / ?

6) have / questions / any / Do / you / ?

B) Complete the following extract from a job interview with the sentences and questions you've formed before:

Interviewer: (1) _____

Interviewee: Good morning, pleased to meet you, too.

Interviewer: (2) _____

Interviewee: Sure!

Interviewer: (3) _____

Interviewee: I've been working as a junior developer for 18 months in DatalQ. I've been part of a team that has been successful in its projects. We've designed an application for managing stock for a large retail chain. Previously, we had planned and designed an app for measuring specific actions on website traffic.

Interviewer: (4) _____

Interviewee: Not there, but I have been captain of my volleyball team once. It happened when I was in high school. We won 3-0 to St George's team that day. We had never beaten a St George's team before.

Interviewer: Ok, and (5) _____

Interviewee: Well, that would depend on the result of this interview, but I expect to start working for this company and stay here for a long time. I had another interview this week, but I didn't feel comfortable at all. If I don't feel comfortable in a place, I prefer not to stay there. By the end of this year, I will have achieved all the company goals. I've been studying the company profile and it's been made very clear to me what your expectations are. On the strength of this success, I will earn your confidence and in five years I will have started managing my own team.

Interviewer: (6) _____

Interviewee: Yes, I do. In fact I would like to discuss some specifics about the contact.

SPEAKING. Work in small groups.

- Why do you think the interviewee told the interviewer about his leadership in the volleyball team? Is it OK to talk about your achievements even though they are not directly related to the position you are applying for? Would you do the same?
- Is it important to ask questions at the end of a job interview? What do you think the interviewee will ask?

GRAMMAR. Work in groups. Revision.

1. Identify an example of a sentence or clause which contain examples of:

- a) present perfect simple
- b) present perfect continuous
- c) past perfect
- d) simple past
- e) future perfect
- f) passive voice (in which verb tense is it written?)

2. Write four conditional sentences (one of each type) from the point of view of the interviewee.

Type 0:

Type 1:

Type 2:

Type 3:

WATCH A VIDEO.

You're hired - Episode 05

You will see Sarah going for a job interview. How well she did?

VOCABULARY. Match the definitions (a-h) with the vocabulary (1-8).

Vocabulary

1. to diversify
2. picky
3. voluntary work
4. technicalities
5. to pull something off
6. to move on
7. carbon footprint
8. temperament

Definitions

- a. work that is unpaid, often for charity organisations
- b. to leave a job
- c. a prevailing or dominant quality of mind that characterises somebody
- d. to become more varied or make something more varied
- e. precise or technical details in a contract
- f. hard to please or satisfy; knowing what you want or like and not accepting anything else
- g. to do something successfully
- h. the amount of carbon dioxide produced by somebody or something

After you watch

Task 1

Are the sentences true or false? Answer

1. Sarah has some experience of doing voluntary work.
2. Sarah's role at the moment is assistant sales director.
3. Sarah thinks that the new job would involve very different skills from her current job.
4. Sarah wants a new job because she has too much responsibility in her current job.
5. When two of her current staff had problems with each other, Sarah helped improve the situation.
6. Sarah thinks it's useful to diversify business strategies.
7. Sarah won a scholarship to go to university in India.

8. Sarah is interested in environmental issues.

Task 2

Highlight the best answer to the interview questions.

1. Can you tell us a little bit about your current position?
 - a. I worked on some things and I got a bit of experience.
 - b. It's a very boring job, and the manager is cruel and unfair.
 - c. I'm assistant sales director for a chain of language schools. My main duties are ...
2. Why are you thinking of moving on?
 - a. My current company is too small for me.
 - b. I've come as far as I can in my current position. I would love to take on some more responsibility.
 - c. I'm looking for a job with a higher salary.
3. Can you give us an example of when you had to deal with a particularly difficult manager?
 - a. I haven't had one. They're all easy to manage.
 - b. A couple of years ago I was working with a guy. He was lazy so I told him to get out.
 - c. I had two people working for me who didn't get along. They had no shared values, different temperaments and argued a lot. So, I sat down with them and we talked about their differences and their problems. After that, things went much more smoothly.
4. How would you deal with a rapidly changing and uncertain global market?
 - a. I'd say to stay flexible and to diversify markets and sales strategies.
 - b. Don't change anything at all. Just continue on as normal, regardless.
 - c. I'm not sure.
5. Any plans to learn other languages?
 - a. Not really.
 - b. I'd love to try!
 - c. Why?
6. Do you have any questions for us?
 - a. So, have I got the job?
 - b. Yes, about your ethical policy and your carbon footprint. I was wondering whether you are planning to reduce your carbon footprint and whether all your products are ethically sourced.
 - c. No.

SPEAKING. Work in pairs and then tell the class. What were Sarah's good and bad points?

You're hired - Episode 09

Before accepting the job offer, Sarah negotiates a few terms of her contract.

Preparation task

Match the definitions (a-h) with the vocabulary (1-8).

Vocabulary

1. stock options
2. to compensate
3. specifics
4. provisionally
5. fire away
6. a pay scale
7. a fixed salary
8. a bonus

Definitions

- a. an informal phrase used to invite someone to ask a question
- b. precise details
- c. a pay system where how much you earn depends on how much experience you have, your qualifications or other conditions
- d. a salary that never changes
- e. to pay somebody for work they have done
- f. money given to an employee in addition to normal pay, especially for reaching a performance target
- g. the opportunity to buy shares in a company
- h. temporarily or conditionally

Task 1

Are the sentences true or false? Answer

1. In this episode, Sarah starts her new job.
2. Sarah wants to discuss some details of the contract.
3. Sarah wants a bonus for meeting her sales targets.
4. It is certain that Sarah will be offered the bonus scheme.
5. Marcia offers Sarah more holiday time.
6. Sarah is a good negotiator.

Task 2

Put the words in the correct order to make sentences.

1. if I can. / ask / a few more / wanted to / questions, / I just

2. I sign the contract / Before / one or two / there were / I wanted to talk about. / specifics

3. adding certain bonuses. / a higher level on the pay scale, / asking for / I thought this could be compensated by / Instead of

4. If / I would be looking for / I make certain sales targets, / or even break them, / or stock options in the company. / a good cash payment

5. there are / for a high-pressure job, / 25 days' holiday. / I also see that / though. / It's not very much

6. my start date. / if / delay / I / we could / was wondering

SPEAKING. Is it common in your area of work to negotiate the terms of a new job, as Sarah did?

Unit N° 7 - Technical Report

Technical Report

A technical report is a document that clearly and in some detail, without being too lengthy, presents the development or result of an investigation or project, enabling a quick understanding of a complex situation.

A good report in English follows a logical structure that facilitates reading and comprehension. Regardless of the type of report you are writing, these are the essential parts you should include:

1. **Title, Date, Problem, and Executive Summary**

It should be clear and reflect its content. Additionally, many reports begin with an executive summary, which is a brief overview that explicitly states the problem to be solved based on the options provided.

Example:

The Impact of Remote Work on Productivity

The executive summary summarizes the most important points of the report in one or two paragraphs, including the main findings and recommendations.

2. **Introduction**

Establishes the purpose and context. This is where you explain the "why" behind your report, what problem you are addressing, and what you hope to achieve.

Example:

The aim of this report is to analyze the effects of remote work on employee productivity during the pandemic.

3. **Development: Body of the Report**

Presents detailed information. It is important to structure this section into different subsections with clear titles that reflect the content. Use headings to divide your text into thematic sections, making the report easier to follow.

For example, if you are writing a report on the effects of remote work, you could divide the body of the report into sections such as:

- Impact on employee productivity
 - Challenges faced
 - Recommendations for improvement
4. Be sure to use connectors like "furthermore," "however," and "in addition" to ensure that ideas flow coherently between paragraphs. The language used in reports should be formal, clear, and direct. This is not the time to be poetic or use colloquial phrases.

Some key terms and expressions that will be useful when writing a report in English are:

- **Expressions to introduce objectives or purposes**
 - The aim of this report is...
 - The purpose of this report is...
 - These phrases are ideal for opening your introduction and giving a clear idea of why you are writing the report.
 - **Vocabulary to present information**
 - According to the data...
 - The findings indicate that...
 - The research suggests...
 - These expressions are very useful for discussing facts or figures in a professional and objective manner.
 - **Phrases to make recommendations**
 - It is recommended that...
 - In order to improve...
 - It is advised to...
5. Recommendations usually appear in the final section of a report, and these phrases will allow you to propose actions or solutions clearly.
6. **Conclusion**

The conclusion should not introduce new information but rather consolidate what you have already presented in the body of the report.

Example:

In conclusion, the analysis suggests that remote work has both positive and negative effects on employee productivity. It is recommended that companies implement strategies to improve communication and provide adequate support for remote workers.

To be avoided:

- Inappropriate use of verb tenses: Ensure that verb tenses are consistent throughout the report. If you are talking about past events, use the past simple, but if you are making recommendations or projections, the future will be your best option.
- Colloquial or informal phrases: A report is a formal document, so you should avoid informal phrases like "gonna" or "wanna." Maintain a professional tone at all times.
- Lack of connectors: Connectors are essential for your report to flow. Use words like "therefore," "in addition," "on the other hand" to guide the reader through your ideas.

7. Writing reports in English may seem like a difficult task at first, but with the right structure and key vocabulary, you can do it effectively and professionally.
Always remember to maintain formal language, use connectors to make your ideas flow, and divide your report into clear sections. By following these steps, you will be ready to write reports in English that are not only informative but also easy to understand and well-structured.

Technical Report: Wireless Network Infrastructure Upgrade

Date: October 26, 2023

To: Network Operations Manager

From: IT Infrastructure Team

Executive Summary:

This report details the evaluation and recommendation for upgrading the existing wireless network infrastructure to address increasing bandwidth demands and improve overall network performance. We assessed several Wi-Fi 6E access point models, considering factors such as throughput, security features, and cost-effectiveness. Based on our analysis, we recommend the [Specific Model, e.g., "Ubiquiti UniFi 6E Enterprise"] for deployment, which offers superior performance and future-proofing capabilities.

1. Introduction

The purpose of this report is to analyze the current limitations of our wireless network and recommend a suitable upgrade path to meet the growing demands of our user base. The existing Wi-Fi 5 infrastructure is experiencing performance bottlenecks due to increased device density and bandwidth-intensive applications. This upgrade aims to enhance network reliability, speed, and security.

2. Evaluation Criteria

The following criteria were used to evaluate potential Wi-Fi 6E access point models:

- **Throughput:** Maximum data transfer speeds and real-world performance.
- **Security:** Support for WPA3 encryption and advanced security features.
- **Device Capacity:** Ability to handle a high density of connected devices.
- **Power over Ethernet (PoE):** Compatibility with existing PoE infrastructure.
- **Management and Monitoring:** Ease of configuration and network management tools.
- **Cost-Effectiveness:** Overall value proposition, including upfront and ongoing costs.

3. Comparative Analysis

We evaluated the following Wi-Fi 6E access point models:

- [Model A, e.g., "Cisco Catalyst 9136"]
- [Model B, e.g., "Aruba 655 Series"]
- [Model C, e.g., "Ubiquiti UniFi 6E Enterprise"]
- [Model D, e.g., "TP-Link Omada EAP670"]

3.1. Throughput and Performance:

- Model C demonstrated the highest real-world throughput in our testing, particularly in congested environments.
- Models A and B also provided excellent performance, but at a higher cost.
- Model D had acceptable speeds, but did not perform as well under high load.

3.2. Security Features:

- All models support WPA3 encryption.
- Models A and B offer advanced security features, such as intrusion detection and prevention, which are valuable for high-security environments.
- Model C and D also offer robust security, but with less advanced features.

3.3. Device Capacity and PoE:

- All models support high device density and PoE+.
- Model C had the best performance under high density testing.

3.4. Management and Monitoring:

- Models A and B offer comprehensive management platforms with advanced analytics.
- Model C offers a user-friendly, cloud-based management interface.
- Model D offers a good management interface, but less advanced features.

3.5. Cost-Effectiveness:

- Model C provides the best balance of performance and cost.
- Models A and B are significantly more expensive.
- Model D is less expensive, but has lower performance.

4. Recommendation

Based on our analysis, we recommend the [Specific Model, e.g., "Ubiquiti UniFi 6E Enterprise"] for upgrading our wireless network infrastructure. This model offers:

- Excellent throughput and performance.
- Robust security features.
- High device capacity.

- User-friendly management.
- Very good cost effectiveness.

5. Implementation Plan

- Phase 1: Conduct a site survey to determine optimal access point placement.
- Phase 2: Procure and configure the new access points.
- Phase 3: Deploy the new infrastructure in a phased approach, minimizing disruption.
- Phase 4: Perform post-deployment testing and optimization.
- Phase 5: Decommission the old access points.

6. Conclusion

The upgrade to Wi-Fi 6E will significantly improve our wireless network performance, providing a more reliable and efficient experience for our users. Implementing the recommended model will ensure that our network can meet the demands of current and future technologies.

Sincerely,

IT Infrastructure Team

Speaking Unit N°1

Student B

JavaScript was created by Brendan Eich at Netscape Communications in 1995. Initially named Mocha, then LiveScript, it was later renamed JavaScript. The name does not imply a direct relationship with Java; the naming choice was influenced by the popularity of Java at the time. JavaScript is primarily known as a scripting language for web browsers, enabling interactive and dynamic web pages. While not inherently visual in the same way as Visual Basic, JavaScript frameworks and libraries like React, Angular, and Vue.js allow for the creation of rich, interactive user interfaces. Instead of a drag-and-drop environment, developers use code to manipulate the Document Object Model (DOM) and create interface elements like buttons, forms, and animations. JavaScript's flexibility extends beyond the browser, allowing for server-side development with Node.js and the creation of mobile applications with frameworks like React Native.